

System of First Order ODEs

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Second Order ODEs

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$$y'' + a_1(x) \cdot y' + a_0(x) \cdot y = b(x)$$

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$$y_1 = y$$

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Transformation of nth Order ODEs

$$\frac{dy_1}{dx} = f_1(x, y_1, y_2, \dots, y_n)$$

$$\frac{dy_2}{dx} = f_2(x, y_1, y_2, \dots, y_n)$$

$\vdots$

$$\frac{dy_n}{dx} = f_n(x, y_1, y_2, \dots, y_n)$$

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Alternative Expression

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Alternative Expression

$$y' = z = f_1(x, y, z)$$

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$$k_{1,1} = f_1(x_i, y_i, z_i)$$

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The Midpoint Method

$$k_{1,1} = f_1(x_i, y_i, z_i)$$

$$k_{2,1} = f_1(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_{1,1}, z_i + \frac{h}{2}k_{1,2})$$

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$$y_{i+1} = y_i + h \cdot k_{2,1}$$

$$k_{1,2} = f_2(x_i, y_i, z_i)$$

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Fourth Order Explicit Runge-Kutta Method

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Fourth Order Explicit Runge-Kutta Method

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$$y_{i+1} = y_i + h \cdot k_{2,1}$$

$$k_{1,2} = f_2(x_i, y_i, z_i)$$

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$$z_{i+1} = z_i + h \cdot k_{2,2}$$

Fourth Order Explicit Runge-Kutta Method

$$k_{1,1} = f_1(x_i, y_i, z_i)$$

$$k_{2,1} = f_1(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_{1,1}, z_i + \frac{h}{2}k_{1,2})$$

$$k_{3,1} = f_1(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_{2,1}, z_i + \frac{h}{2}k_{2,2})$$

$$k_{4,1} = f_1(x_i + h, y_i + hk_{3,1}, z_i + hk_{3,2})$$

$$y_{i+1} = y_i + \frac{h}{6} \cdot (k_{1,1} + 2k_{2,1} + 2k_{3,1} + k_{4,1})$$

$$k_{1,2} = f_2(x_i, y_i, z_i)$$

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System of First Order ODEs

Example 07-09: Approximate the numerical solution of the following initial value problem using **the midpoint method** for $0 \leq x \leq 2$ with step size $h = 0.1$.

$$y'' + 2y' + 4y = 0, y(0) = 2, y'(0) = 0; \quad 0 \leq x \leq 2$$

System of First Order ODEs

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$$y'' + 2y' + 4y = 0, y(0) = 2, y'(0) = 0; \quad 0 \leq x \leq 2 \quad \text{Exact Solution: } y(x) = 2e^{-x} \cdot \left[\cos(\sqrt{3}x) + \frac{1}{\sqrt{3}} \sin(\sqrt{3}x) \right]$$

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$$y' = z = f_1(x, y, z)$$

$$z' = -4y - 2y'$$

$$= -4y - 2z = f_2(x, y, z)$$

System of First Order ODEs

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System of First Order ODEs

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System of First Order ODEs

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System of First Order ODEs

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System of First Order ODEs

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System of First Order ODEs

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System of First Order ODEs

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System of First Order ODEs

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$$k_{1,1} = f_1(x_0, y_0, z_0) = z_0 = 0$$

$$k_{2,1} = f_1\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_{1,1}, z_i + \frac{h}{2}k_{1,2}\right)$$

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System of First Order ODEs

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System of First Order ODEs

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System of First Order ODEs

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System of First Order ODEs

Example 07-09: Approximate the numerical solution of the following initial value problem using **the midpoint method** for $0 \leq x \leq 2$ with step size $h = 0.1$.

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$$\begin{aligned} k_{2,1} &= f_1\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}k_{1,1}, z_0 + \frac{h}{2}k_{1,2}\right) = z_0 + \frac{h}{2}k_{1,2} \\ &= 0 + 0.05 \cdot (-8) = -0.4 \end{aligned}$$

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$$y'' + 2y' + 4y = 0, y(0) = 2, y'(0) = 0; \quad 0 \leq x \leq 2 \quad \text{Exact Solution: } y(x) = 2e^{-x} \cdot \left[\cos(\sqrt{3}x) + \frac{1}{\sqrt{3}} \sin(\sqrt{3}x) \right]$$

$$\left. \begin{aligned} y' &= z = f_1(x, y, z), & y(0) &= 2 \\ z' &= -4y - 2y' & \\ &= -4y - 2z = f_2(x, y, z), & z(0) &= 0 \end{aligned} \right\} \quad 0 \leq x \leq 2, N = \frac{2-0}{0.1} = 20, x_0 = 0, y_0 = 2, z_0 = 0$$

The Midpoint Method:

$$k_{1,1} = f_1(x_i, y_i, z_i)$$

$$k_{2,1} = f_1\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_{1,1}, z_i + \frac{h}{2}k_{1,2}\right)$$

$$y_{i+1} = y_i + h \cdot k_{2,1}$$

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$$\begin{aligned} k_{2,1} &= f_1\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}k_{1,1}, z_0 + \frac{h}{2}k_{1,2}\right) = z_0 + \frac{h}{2}k_{1,2} \\ &= 0 + 0.05 \cdot (-8) = -0.4 \end{aligned}$$

$$\begin{aligned} k_{2,2} &= f_2\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}k_{1,1}, z_0 + \frac{h}{2}k_{1,2}\right) \\ &= -4 \cdot \left(y_0 + \frac{h}{2}k_{1,1}\right) - 2 \cdot \left(z_0 + \frac{h}{2}k_{1,2}\right) \end{aligned}$$

System of First Order ODEs

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System of First Order ODEs

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$$y_1 = y_0 + h \cdot k_{2,1} = 2 + 0.1 \cdot (-0.4) = 1.96$$

System of First Order ODEs

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$$z_1 = z_0 + h \cdot k_{2,2} = 0 + 0.1 \cdot (-7.2) = -0.72$$